req SysML Requirement Diagram - Autonomous Arduino Car

«requirement»
REQ-00: Autonomous Arduino Car
Text: The car shall autonomously follow a line and stop when obstacle is detected.

«requirement»
REQ-01: Line Following
(from REQ-00: Autonomous Arduino Car)
Text: The car shall autonomously follow a line.

«requirement»
REQ-02: Obstacle Detection and Response
Text: The car shall detect obstacles and respond safely.

<<satisfy>>
<<satisfy>>
<<satisfy>>
<<satisfy>>

«requirement»
FR-00: Functional Requirements
Text: The system shall provide all required functions for line following and obstacle detection.

«requirement»
NFR-00: Non-Functional Requirements
Text: The system shall meet quality, safety, reliability, and hardware constraints during operation.

«requirement»
FR-01: Line Detection
Text: The car shall detect the track line using two front IR sensors.

«requirement»
FR-02: Line Following Control
Text: The car shall follow the detected line and move forward, left or right by controlling motor movement.

«requirement»
FR-05: Obstacle Detection
Text: The car shall detect obstacles using two front ultrasonic sensors.

«requirement»
FR-07: Obstacle Stop
Text: The car shall stop when an obstacle is within the defined safety distance.

«requirement»
FR-08: Motor Control
Text: The car shall control two DC motors using a motor driver.

<<refine>>
<<refine>>
<<refine>>
<<refine>>
<<refine>>

«requirement»
FR-03: Direction Correction
Text: The car shall correct its direction based on left and right IR sensor readings.

«requirement»
FR-04: Line Lost Handling
Text: The car shall stop or enter a safe state when the line is lost.

«requirement»
FR-06: Distance Measurement
Text: The car shall calculate obstacle distance using ultrasonic echo timing.

<<deriveReq>>
<<deriveReq>>

«requirement»
DR-01: IR Sensor Placement
Text: The two IR sensors shall be mounted at the front of the car and aligned with the track line.

«requirement»
DR-02: Ultrasonic Sensor Placement
Text: The two ultrasonic sensors shall be mounted at the front of the car to cover the obstacle detection area.

«requirement»
NFR-01: Real-Time Response
Text: The car shall respond quickly enough to sensor changes to maintain stable movement.

«requirement»
NFR-02: Safety
Text: The car shall prioritize stopping when an unsafe condition is detected.

«requirement»
NFR-03: Reliability
Text: The car shall operate consistently during repeated test runs.

«requirement»
NFR-04: Mechanical Stability
Text: Components shall remain securely mounted during operation.

«requirement»
NFR-05: Power Compatibility
Text: The system shall operate within voltage and current limits.

<<deriveReq>>
<<deriveReq>>
<<deriveReq>>

«requirement»
DR-03: Motor Driver Interface
Text: The controller shall send PWM and direction signals to the motor driver.

«requirement»
DR-04: Safe Stop Command
Text: The motor driver shall stop both motors when a stop command is received.

«requirement»
DR-05: Stable Component Mounting
Text: The chassis shall provide secure mounting points for sensors, motors, controller, battery, and wiring.